

Problem Setup

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Problem Setup

We are given the algebraic equation:

$$x^2 + 2u + 1 = 0$$

The objective is to express one variable in terms of the other. Depending on the context of the problem, we typically solve for u as a function of x , or solve for x in terms of u . We will provide both perspectives to ensure a complete solution.

Strategy and Key Principles

To solve this equation, we apply standard algebraic manipulation techniques:

1. Isolation of terms: Move all terms not containing the target variable to the opposite side of the equation.
2. Division/Multiplication: Divide or multiply by coefficients to completely isolate the target variable.
3. Radical extraction: When solving for a variable squared, apply the square root property while accounting for both positive and negative branches.

Detailed Solution

First, let us solve for u in terms of x . We begin with the original equation:

$$x^2 + 2u + 1 = 0$$

Subtract x^2 and 1 from both sides of the equation:

$$2u = -x^2 - 1$$

Divide both sides by 2:

$$u = \frac{-x^2 - 1}{2}$$
$$u = -\frac{x^2 + 1}{2}$$

Next, let us solve for x in terms of u . We begin again with the original equation:

$$x^2 + 2u + 1 = 0$$

Subtract $2u$ and 1 from both sides of the equation:

$$x^2 = -2u - 1$$

Take the square root of both sides, remembering to include the $\pm$ symbol to account for both real (if $-2u - 1 \geq 0$) or complex branches:

$$x = \pm\sqrt{-2u - 1}$$

Final Answer

Depending on which variable is treated as the dependent variable, the solutions are:

$$u = -\frac{x^2 + 1}{2} \quad \text{or} \quad x = \pm\sqrt{-2u - 1}$$

$$\boxed{u = -\frac{x^2 + 1}{2}}$$

Key Insights

- **Dual Perspectives:** Depending on the broader mathematical context (such as implicit curves or multivariable functions), this equation can define u explicitly as a function of x , or x as a multi-valued function of u .
- **Domain Restrictions:** If we consider real numbers only, solving for x requires $-2u - 1 \geq 0$, which restricts the domain to $u \leq -\frac{1}{2}$. Conversely, solving for u in terms of real x yields $u \leq -\frac{1}{2}$ automatically because $x^2 + 1 \geq 1$.
- **Sign Conventions:** Failing to include the $\pm$ symbol when taking the square root to solve for x is a common error that discards half of the solution space.